



**AMERICAN
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Wind Turbine Airfoil Comparison for Residential Applications in Lebanon

MECH 683 – Wind Turbines: Design & Application

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Submitted to: Professor Samer Salloum

February 6, 2026

Abstract

This project investigates the aerodynamic efficiency and power-generating capability of two selected airfoils for horizontal-axis wind turbine applications in Lebanon. Using Q-Blade, the airfoils were modeled, simulated, and analyzed under local wind conditions to assess glide ratio, pressure distribution, and overall turbine performance. The study aims to determine the most suitable airfoil profile for small-scale residential wind turbine installations in Lebanon. We have chosen NACA-4412 and NACA-6409 as our airfoils, from our finding the NACA-4412 performed better in terms of power generation at normal conditions and the NACA-6409 performed better under optimized conditions, with the best of our abilities we made this report come into fruition.

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1 Introduction and Background

1.1 introduction

Lebanon's present energy situation is characterized by widespread, protracted daily black-outs due to an enormous, unmet demand for electricity. A large portion of the primary energy supply comes from imported fossil fuels, and the power infrastructure is severely out-of-date and inefficient due to significant technical losses. Residents and businesses now rely on costly and extremely polluting private diesel generators as a result of the on-going supply shortage. Furthermore, the sector is plagued by governance issues and a lack of investment in updating the transmission network, which precludes the construction of a steady, reliable power supply for the country .

This is where the Renewable energy can have an advantage

- **Diversification of Generation:** Adding additional, domestically produced capacity through wind power is essential for reducing dependency on imported fuel oil and diesel and enhancing energy security.
- **Reduces reliance on diesel generators:** Diesel generators are the primary backup power source during blackouts and a significant contributor to greenhouse gas emissions and air pollution. Renewable systems, especially wind and solar can replace a portion of this generator use, benefiting public health and environmental quality (UNDP Lebanon, 2022).
- **lessens reliance on imported fossil fuels:** at the moment fossil fuels leave Lebanon very susceptible to fluctuations in world prices and supplies. Expanding renewable energy provides a more dependable, locally supplied option that promotes national energy security (World Bank, 2021).
- **Provides substantial business and job prospects:** Since the renewable energy sector requires trained workers for installation, maintenance, system design, and technical support. This development can promote local employment markets and help long-term economic prosperity (IRENA, 2020) .

1.2 Background

As stated above, renewable energy plays significant roles in different sectors, whether it is in diversifying the energy sources in Lebanon, Optimizing energy cost in Lebanon or in providing new jobs and opportunities. In Lebanon solar electricity has been a reliable source for the Lebanese people for the last few years and this is mainly due to the increased price of fuel and diesel, but there is several renewable sources that can be exploited particularly wind energy can be an efficient and steady source mainly due to Lebanon's unique and natural conditions such as the diversity of locations (urban and rural settings, Mountains for ex) and the possession of favorable wind condition.

For a better visualization, tables representing Lebanon's mean wind speeds across all the months are presented, these tables have extracted from: https://files.acquia.undp.org/public/migration/lb/National_Wind_Atlas_report.pdf

Table 1: Monthly Mean Wind Speeds (m/s) Across Lebanese Regions at 10m elevation

Region	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Mean
Dahr-el-Baidar	6.0	6.1	5.9	5.5	5.8	5.0	5.0	4.7	4.3	4.9	7.4	5.4	5.4
Les Cèdres	4.4	4.4	4.4	4.4	3.5	3.3	3.5	3.5	3.3	3.6	9.6	3.9	3.9
Abdé	2.4	2.4	2.4	2.4	2.7	2.3	2.3	2.3	2.3	2.5	1.1	2.6	2.6
Klaaiat	3.9	5.1	5.1	4.5	3.5	3.8	3.2	3.2	2.8	3.9	1.6	4.2	4.2
Akkar	3.9	3.9	3.4	3.2	3.1	2.9	2.8	2.8	2.6	2.8	8.8	3.0	3.0
Sour	1.9	2.4	2.4	2.2	2.2	1.9	1.6	1.6	1.6	2.2	1.1	2.0	2.0
Tripoli	3.9	3.9	3.4	3.2	3.1	2.9	2.8	2.8	2.6	2.8	8.8	3.0	3.0
Beirut – Golf	3.0	3.0	3.0	3.0	2.6	2.5	2.5	2.5	2.2	2.2	1.1	2.8	2.8
Beirut Airport	3.7	3.5	3.5	3.4	3.1	3.0	3.0	3.0	2.7	2.8	8.8	3.2	3.2

Table 2: Monthly Mean Wind Speeds (m/s) Across Lebanese Regions at 10m elevation

Region	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Mean
Zahrani	4.0	4.6	4.4	4.1	3.7	3.7	3.6	3.7	3.6	3.7	4.3	4.0	4.0
Marjayoun	5.9	5.8	5.4	7.3	8.9	7.1	6.4	6.8	7.4	6.8	0.7	-2.0	–
Hermel	3.2	3.2	3.5	3.4	3.8	2.8	2.8	2.7	2.6	2.5	1.4	3.1	3.1
Faqra	2.7	2.5	2.9	3.3	3.6	1.8	1.8	1.8	1.6	1.7	1.1	2.4	2.4
Qaraoun	4.0	4.0	3.7	3.6	3.8	3.6	3.0	3.0	2.7	3.1	2.1	1.8	3.8
Rayak	3.7	3.8	3.4	3.8	3.7	3.5	3.4	3.5	3.0	3.6	2.1	3.6	3.6
Zahleh	2.6	2.7	2.7	2.9	2.6	2.6	2.5	2.6	2.9	2.7	2.5	2.6	2.6
Bayssour	4.2	4.0	4.2	4.0	4.0	3.6	3.2	3.2	3.4	3.2	7.7	3.4	3.4

According to the IEC 61400-1 Standard wind cites are classified under 3 types:

- **Class I:** Low wind cites (0-7.5) m/s
- **Class II:** Moderate wind cites (7.5-8.5) m/s
- **Class III:** Very windy Cites (8.5-10) m/s

These values are measured at the hub-height.

In order to estimate to which class/es Lebanon belongs to, we are going to take a hub height = 80 m used for Wind Farms, According to the IEA large onshore Wind-Turbines (Wind-Turbines installed on lands, hills, fields ...) should have a hub-height ranging between (80-120) m. From the Tables above, the highest mean speed at hub-height of 10m is at Dahr-el-Baidar at 5.4m/s and the lowest mean speed is at Sour at 2m/s.

$$\text{Dahr el-Baidar: } z_0 = 0.05 \text{ m} \quad ; \quad \text{Sour: } z_0 = 0.2 \text{ m}$$

Using this correlation:

$$U(z) = U(10) \frac{\ln\left(\frac{z}{z_0}\right)}{\ln\left(\frac{10}{z_0}\right)}$$

We find the following:

$$U_{\text{Dahr-el-Baidar}} = 7.52 \text{ m/s} \quad ; \quad U_{\text{Sour}} = 3.063 \text{ m/s}$$

These findings show that Lebanon belongs to Class I and Class II (Low-Moderate wind cites), from our findings we consider Dahr-el-Baidar to be a valid option view that it has the highest mean wind speed at the selected hub height and that it has vast and wide spaces to handle large wind-turbines.



Figure 1: Figure showing the vast horizons of Dah-el-Baidar

2 Airfoil Selection

2.1 Criteria for Selection

The Wind-Turbine will be implemented in a moderate wind city (Dahr-el-Baidar), most sources suggests that airfoils for moderate wind cities are:

- NACA-4412 and NACA-6409 (Selig et al, 1997) and (Hansen et al, 2015)
- NACA-4415 and NACA-23012 (Spera et al, 2009)
- NACA-4412 and NACA-4415 (kumar, R et al, 2020)

Airfoil	NACA-4412	NACA-4415	NACA-6409	NACA-23012
Max camber	0.04c	0.04c	0.06c	0.15c
Location from leading edge	0.4c	0.4c	0.4c	0.15c
Maximum thickness	0.12c	0.15c	0.09c	0.12c
Lift coefficient (design)	1.00	0.95	1.15	0.30

Table 3: Geometric and aerodynamic properties of selected NACA airfoils.

To ensure a better comparison we have generated cl/cd plots for these airfoils, and out of these 4 we are going to choose 2 for our potential wind turbine model.

To do this analysis we have selected several parameters:

- $U = 7.52$ m/s
- $\rho = 1.2$ kg/m³
- $c = 1$ m
- $\mu = 1.78 \times 10^{-5}$ Pa·s

$$Re = \frac{\rho U c}{\mu} = \frac{(1.2)(7.52)(1)}{1.78 \times 10^{-5}} = 506966$$

.

Once the Reynolds number is found, we can start with our analysis

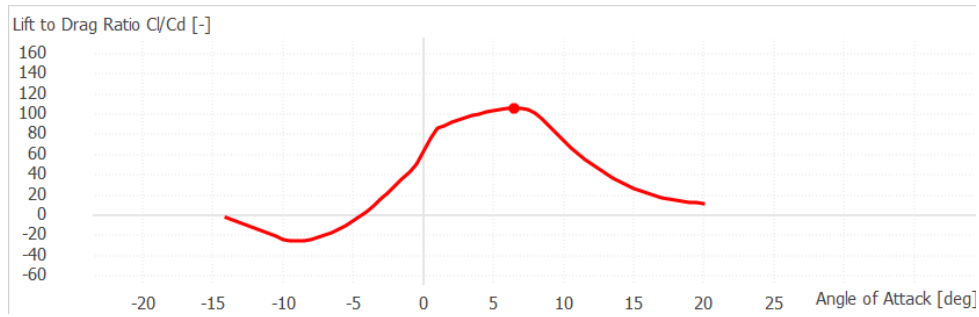


Figure 2: Figure showing the C_l/C_d ratio of the NACA-4412 airfoil

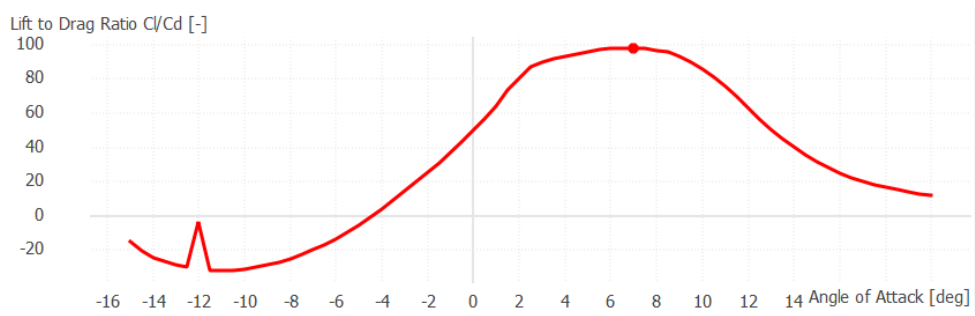


Figure 3: Figure showing the C_l/C_d ratio of the NACA-4415 airfoil

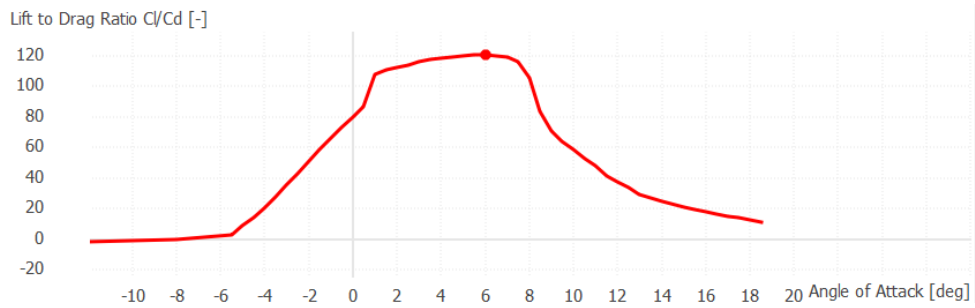


Figure 4: Figure showing the C_l/C_d ratio of the NACA-6409 airfoil

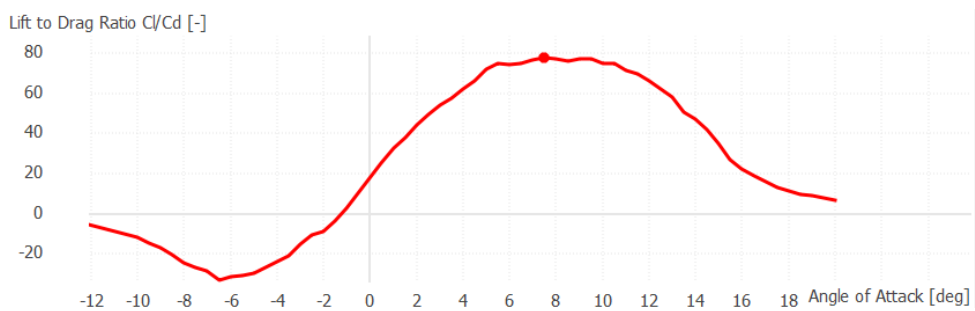


Figure 5: Figure showing the C_l/C_d ratio of the NACA-23012 airfoil

We have combined all of the results in a Table.

Airfoil	NACA-4412	NACA-4415	NACA-6409	NACA-23012
Cl/Cd	106	98.095	120.4	77.8

Table 4: Maximum Cl/Cd for selected NACA airfoils.

We have decided to choose NACA-4412 and NACA-6409.

NACA-4412:

- Has a decent thickness which provides a good structural length.
- Has the 2nd best Cl/Cd ratio

NACA-6409:

Most aerodynamically efficient out of the 4 but has a smaller thickness compared to the rest.

2.2 Airfoil profiles

This section presents the profile of the airfoils and explains their geometrical terminology.

Because they directly affect the blade's aerodynamic performance, it is crucial to

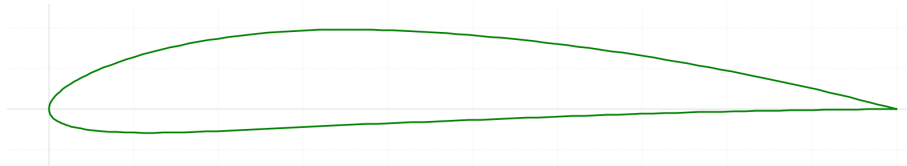


Figure 6: Figure presenting the profile of the NACA-4412 airfoil



Figure 7: Figure presenting the profile of the NACA-6409 airfoil

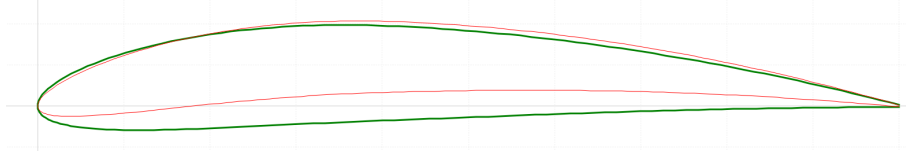


Figure 8: Figure presenting the profile of both airfoils superposed

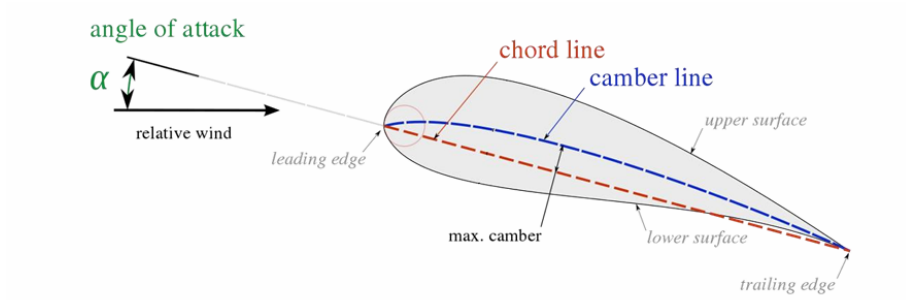


Figure 9: Figure showing the geometrical characteristics of an airfoil

Airfoil	NACA-4412	NACA-6409
Max camber	0.04c	0.06c
Location from the leading edge	0.4c	0.4c
Maximum thickness	0.12c	0.09c

Table 5: Table showing the geometrical properties of the selected airfoils

comprehend the geometrical characteristics of an airfoil, such as camber, thickness, and chord distribution. These factors affect the lift and drag forces, stall characteristics, and structural strength, all of which have an impact on a wind turbine's stability and efficiency. Precise understanding of the airfoil shape guarantees accurate performance forecasting, optimal blade design, and dependable operation under a range of wind conditions.

3 Modeling and Simulation Setup

3.1 Geometric Model of Airfoils

Define:

- **Chord Length:** 3.5 m at the root
- **Thickness-to-chord ratio:** 0.12 (NACA 4412) and 0.09 (NACA-6409)
- **Camber:** 0.04c m (NACA-4412) and 0.06c m (NACA-6409)
- **Camber location:** 0.4c m (NACA-4412) and 0.4c m (NACA-6409)

The generation of lift and drag forces is determined by basic aerodynamic principles that control airfoil performance. Lift is directly impacted by the pressure differential between the top and lower surfaces, which is influenced by the airfoil's curvature (camber) and thickness distribution. Increasing the angle of attack increases lift until stall, at which point flow separation results in an abrupt loss of performance. Surface friction, pressure distribution, and flow separation behavior all affect drag. By encouraging smooth, attached flow and reducing turbulence, efficient airfoils optimize the lift-to-drag ratio, resulting in increased aerodynamic efficiency and overall turbine performance.

3.2 Q-Blade Simulation Setup

Simulation input parameters include:

- Wind speed based on Lebanese regional data : 7.52 m/s at a hub height of 80 m
- Blade length = 45m
- Hub Radius = 5m
- Swept Area = $A = \pi r^2 = A = \pi 50^2 = 7854m^2$
- Number of radial blade segments : 3

Now in order to conduct the simulations we need to find the Reynolds number and Mach's

number. Re was found previously to be 506966, However for Mach's number we are going to apply the following procedure:

- Beirut mean annual temperature (baseline): Climate-Data.org — Beirut average $\approx 18.6^\circ\text{C}$.
- Elevation of Dahr El Baydar: MapLogs — 1506 m.
- Standard atmospheric lapse rate: International Standard Atmosphere (ISA) — 6.5°C per 1000 m.
- The equation and theory of the Ideal-gas speed of sound: $a = \sqrt{\gamma RT/M}$.
- Universal gas constant and molar mass of dry air:

$$R = 8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}, \quad M = 28.9647 \text{ g mol}^{-1} = 0.0289647 \text{ kg mol}^{-1}.$$

Step-by-Step Computation

1. Baseline Temperature

The mean temperature of Beirut is:

$$T_{\text{Beirut}} = 18.6^\circ\text{C}.$$

2. Elevation

Elevation of Dahr El Baydar:

$$h = 1506 \text{ m}.$$

3. Temperature Drop Using the Lapse Rate

Standard lapse rate:

$$\Gamma = 6.5^\circ\text{C}/\text{km}.$$

Altitude difference:

$$1.506 \text{ km}.$$

Temperature decrease:

$$\Delta T = 6.5 \times 1.506 = 9.789^\circ\text{C}.$$

4. Estimated Dahr El Baydar Temperature

$$T_{\text{Dahr}} = 18.6 - 9.789 = 8.811^{\circ}\text{C}.$$

Convert to Kelvin:

$$T = 8.811 + 273.15 = 281.961 \text{ K}.$$

5. Constants for Ideal Gas Speed of Sound

$$\gamma = 1.4, \quad R = 8.314462618 \text{ J mol}^{-1} \text{ K}^{-1}, \quad M = 0.0289647 \text{ kg/mol}.$$

6. Apply the Formula

$$a = \sqrt{\frac{\gamma RT}{M}}.$$

Compute numerator:

$$RT = 8.314462618 \times 281.961 = 2342.988,$$

$$\gamma RT = 1.4 \times 2342.988 = 3279.1832.$$

Divide by molar mass:

$$\frac{\gamma RT}{M} = \frac{3279.1832}{0.0289647} = 113313.6498.$$

Take the square root:

$$a = \sqrt{113313.6498} = 336.6209 \text{ m/s}.$$

Final Value

$$\boxed{a \approx 336.62 \text{ m/s}}$$

$$M_a = \frac{7.52}{336.62} = 0.0223$$

3.2.1 Wind Profile

Table 6: Table showing the Wind Properties in Dahr el-Baidar

Wind Property	Representation	Value
Temperature	T	8.811°C
Atmospheric lapse rate	–	6.5
Specific heat ratio	γ	1.4
Speed of sound	a	336.62 m/s
Altitude	H	1506 m
Molar mass	M	0.0289647 kg/mol
Universal gas constant	R	8.314462618 J/(mol.K)
Dynamic viscosity	μ	$1.78 \times 10^{-5} \text{ kg/(m.s)}$
Air density	ρ	1.2 kg/m^3
Pressure	P	87.9 kPa

3.3 Simulation Objectives

NACA-6409

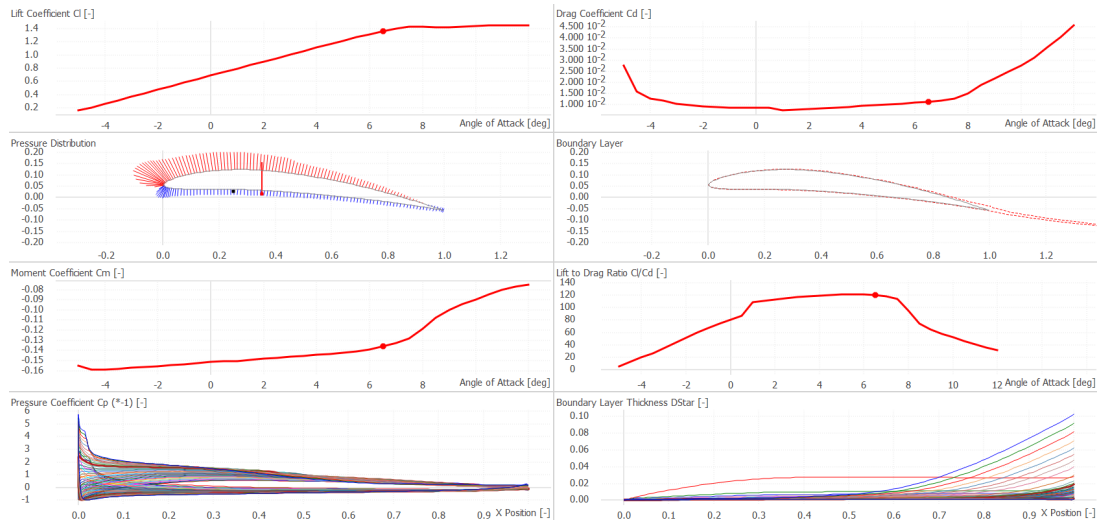


Figure 10: Figure representing the plots of several parameters of the NACA-6409 airfoil

- Max C_l/C_d ratio = 120 , $C_l=1.368$
- Angle of attack = 6.5 degrees

NACA-4412

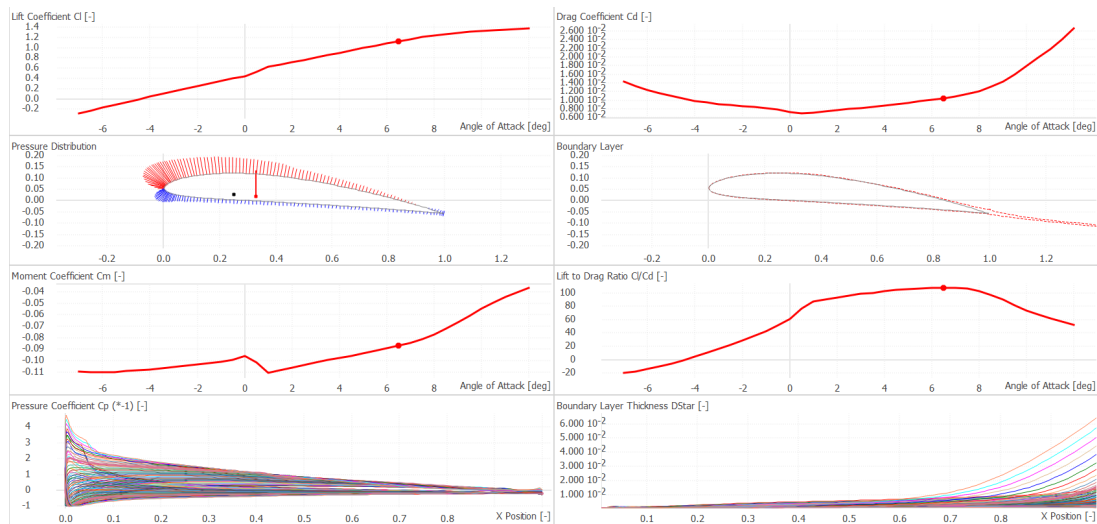


Figure 11: Figure representing the plots of several parameters of the NACA-4412 airfoil

- Max C_l/C_d ratio = 107.7 , $C_l=1.122$
- Angle of attack = 6.5 degrees

4 Performance Analysis

4.1 Glide Ratio (C_l/C_d)

NACA-6409

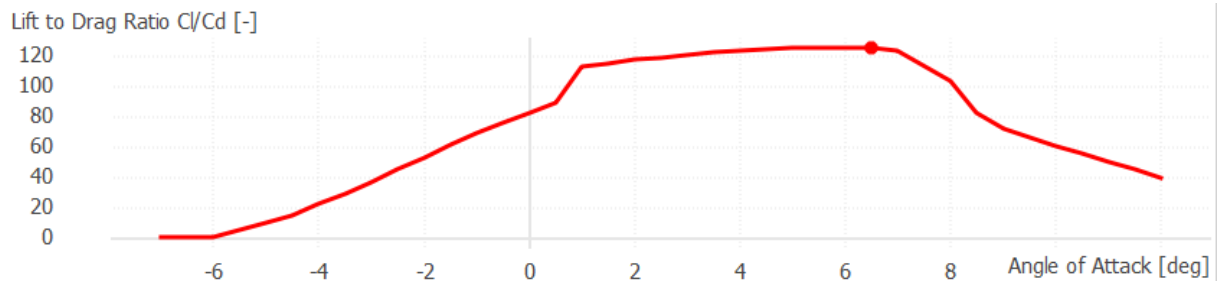


Figure 12: Figure showing the C_l/C_d ratio for the NACA-6409 airfoil across various angles of attack

The glide ratio (C_l/C_d) varies significantly with the angle of attack. At negative angles of attack, the lift produced by the airfoil is relatively small compared to the drag, resulting in a low glide ratio. As the angle of attack increases toward the range of 0° to 4° , the lift increases more rapidly than the drag, causing a pronounced increase in the glide ratio. The curve reaches its maximum value of approximately 120 at an angle of attack of 6.5° , which represents the point of highest aerodynamic efficiency for the airfoil. Beyond this optimal angle, the glide ratio decreases as drag begins to increase faster than lift, reducing the overall aerodynamic performance.

- optimal Angle of Attack at 6.5 degrees
- significant increase from 0 to 4
- C_l/C_d max at 120

NACA-4412

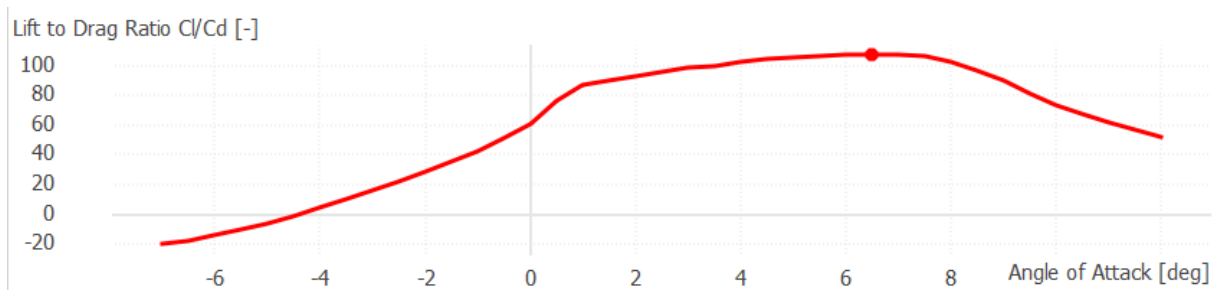


Figure 13: Figure showing the C_l/C_d ratio for the NACA-4412 airfoil across various angles of attack

The glide ratio (C_l/C_d) varies notably with the angle of attack. At negative angles of attack, the lift produced by the airfoil is relatively low compared to the drag, leading to small or negative glide ratio values. As the angle of attack increases toward 0° to 4° , the lift rises more rapidly than the drag, resulting in a pronounced increase in the glide ratio. The curve reaches its maximum value of approximately 107.7 at an angle of attack of 6.5° , which represents the point of highest aerodynamic efficiency for the airfoil. Beyond this optimal angle, the glide ratio decreases as drag begins to grow more quickly than lift, thereby reducing the overall aerodynamic performance.

- optimal Angle of attack at 6.5 degrees
- significant increase from -2 to 4
- C_l/C_d max at 107.7

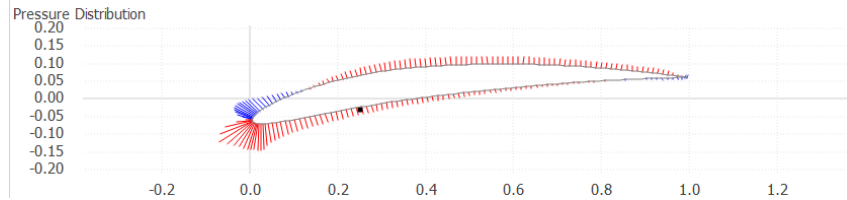
Comparison

At an angle of attack of 6.5 degrees, both airfoils reach their maximum aerodynamic efficiency; nevertheless, the first airfoil achieves a higher peak glide ratio (120) than the second (107.7). This suggests that the first airfoil generates lift more effectively than drag. The first airfoil obviously exhibits superior aerodynamic performance, even though their general trends are comparable.

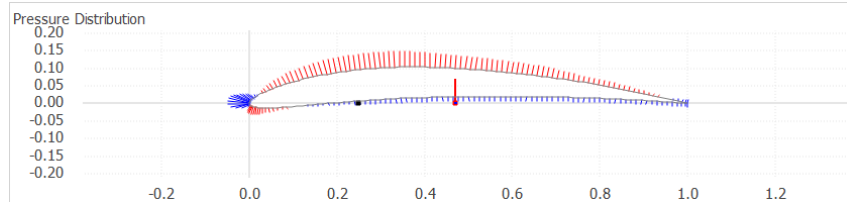
4.2 Pressure Distribution

NACA-6409

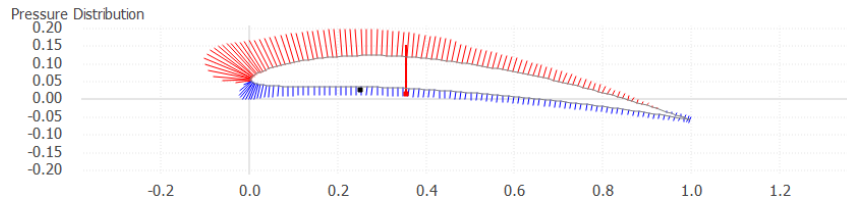
For the Pressure distribution we have chosen 4 points for our analysis, 1st $Cl/Cd = 0$ 2nd 0 degrees 3rd 6.5 degrees last at 12. These points were considered the most critical ones where we had a shift from one phase to another.



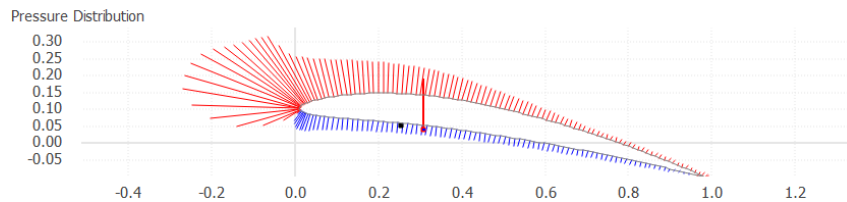
(a) Angle of attack = -7



(b) Angle of attack = 0



(c) Angle of attack = 6.5



(d) Angle of Attack = 12

Figure 14: Combined pressure distribution plots for the NACA-6409 airfoil

- **-7:** small suction peak almost no generation of lift flow is uniform.
- **0:** small suction peak generation of lift the flow is still uniform.
- **6.5:** strong suction peak, lift increased significantly compared to drag.
- **12:** Sharp separation of flow drag increases significantly, high pressure gradient .

NACA-4412

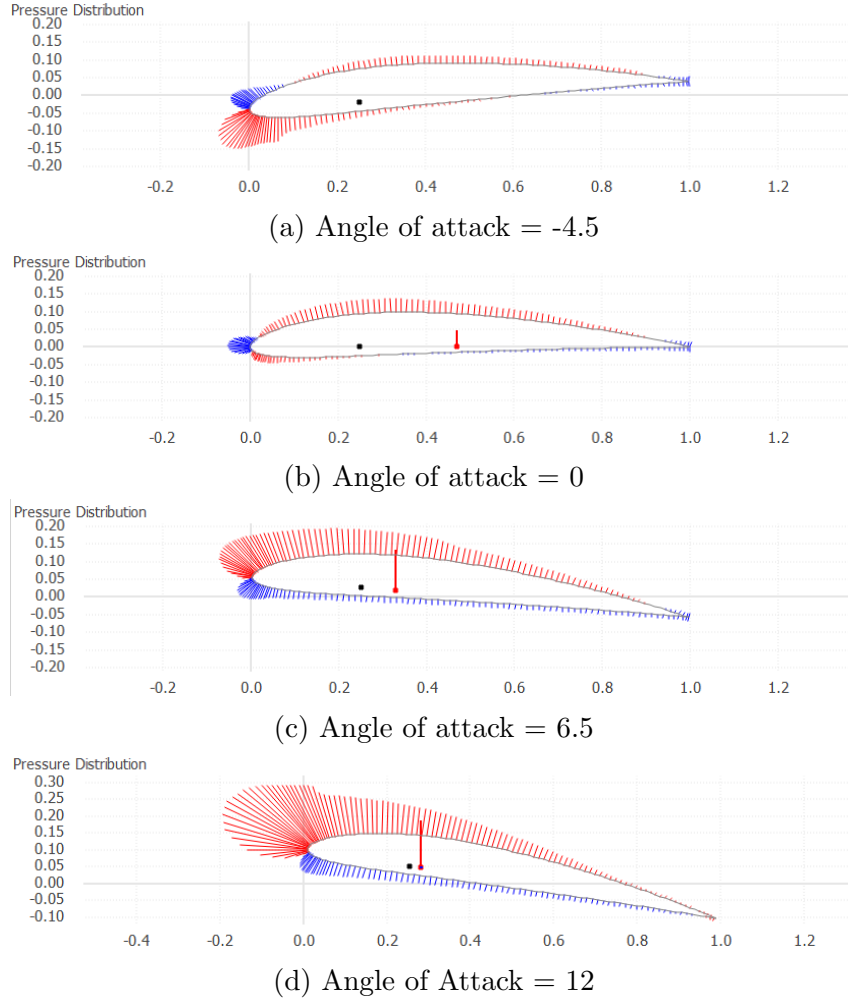


Figure 15: Combined pressure distribution plots for the NACA-4412 airfoil

- **-4.5:** Small suction peak almost no generation of lift flow is uniform.
- **0:** Very small suction peak generation of lift the flow is still uniform.
- **6.5:** Strong suction peak, lift increased significantly compared to drag.
- **12:** Sharp separation of flow drag increases significantly, high pressure gradient .

When comparing the two airfoils at their optimal angles of attack, the NACA 4412 exhibits a stronger suction peak on the upper surface near the leading edge, indicated by the larger and denser red pressure vectors. In contrast, the NACA 6409 shows a weaker suction distribution, with lower peak intensity and a more gradual pressure recovery. This indicates that the NACA 4412 generates greater lift through a more pronounced low-pressure region over the upper surface

Stress analysis

NACA-6409

Table 7: Estimated normal and tangential loading per unit span (NACA 6409, $V = 7.52$ m/s)

r (m)	F_n (N/m)	F_t (N/m)
0.00	0.9	92.8
2.25	35.2	98.1
4.50	242.8	340.8
6.75	361.2	338.3
9.00	488.9	343.1
11.25	624.6	350.1
13.50	765.9	356.9
15.75	911.9	363.4
18.00	1060.8	368.9
20.25	1212.2	373.7
22.50	1365.1	377.8
24.75	1520.2	381.4
27.00	1675.7	384.3
29.25	1832.4	386.7
31.50	1990.5	389.0
33.75	2148.4	390.7
36.00	2307.2	392.2
38.25	2466.3	393.4
40.50	2624.3	394.2
42.75	2785.0	395.1
45.00	2944.3	395.6

By inputting these values in Qblade we get the following: From this figure we can deduce

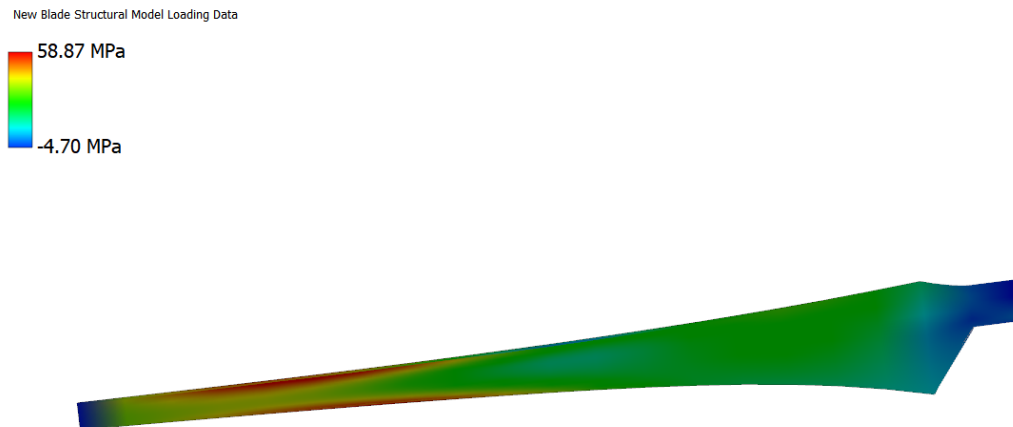


Figure 16: Figure showing the Stress Analysis for the NACA-6409 Airfoil

the following:

- maximum stress = 58.87 MPA located around 0.75 of the chord length.
- minimum stress = 4.7 MPA located at the Tip and at the circular section of the blade.
- the rest of the blade is subject to moderate stresses ranging between (50 and 15) MPA.

Table 8: Normal and tangential loading per unit span for the NACA 4412 blade ($V = 7.52$ m/s)

r (m)	F_n (N/m)	F_t (N/m)
0.00	0.0	0.0
2.25	16.9	47.1
4.50	258.3	362.5
6.75	384.2	359.9
9.00	520.2	365.0
11.25	664.4	372.4
13.50	814.9	379.7
15.75	969.9	386.5
18.00	1128.6	392.5
20.25	1289.7	397.6
22.50	1452.3	401.9
24.75	1617.1	405.7
27.00	1782.8	408.8
29.25	1949.7	411.5
31.50	2116.9	413.7
33.75	2285.4	415.6
36.00	2453.8	417.1
38.25	2623.1	418.4
40.50	2792.4	419.4
42.75	2961.6	420.2
45.00	3132.9	421.0

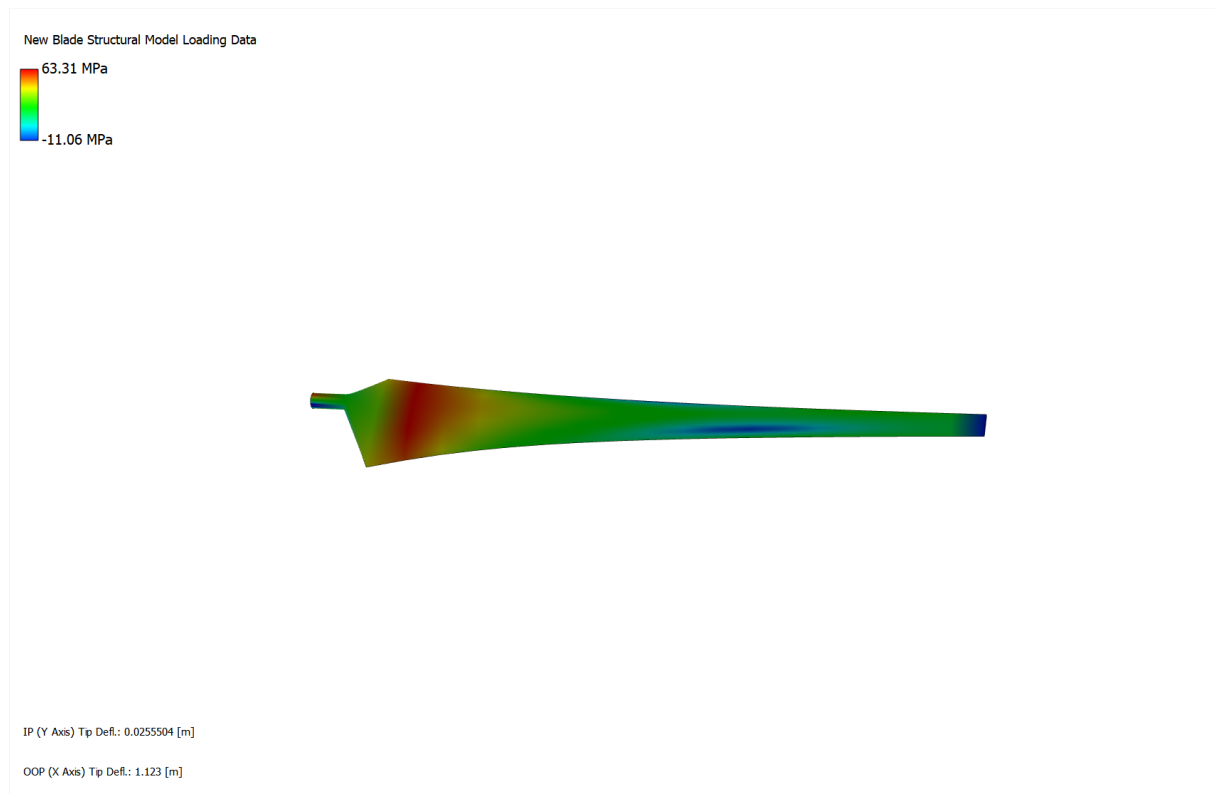


Figure 17: igure showing the Stress Analysis for the NACA-4412 Airfoil

- maximum stress = 63.31 MPA located around 0.75 of the chord length.
- minimum stress = 11.06 MPA located at the Tip and at the circular section of the blade.
- the rest of the blade is subject to moderate stresses ranging between (55 and 20) MPA.

4.3 Power Output

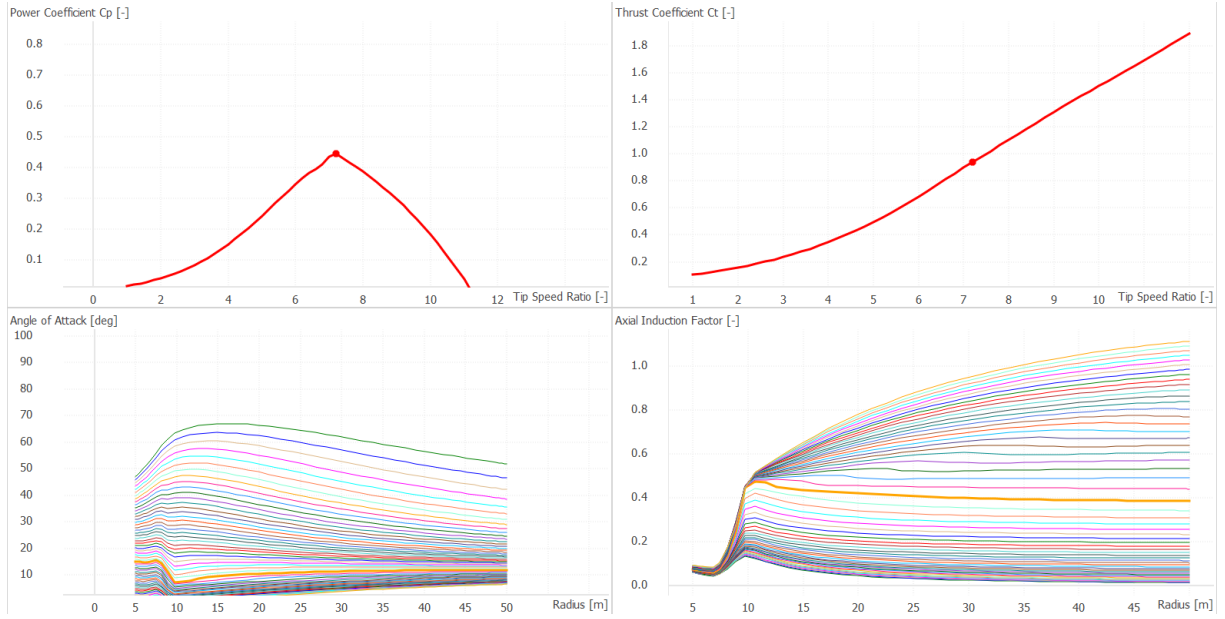


Figure 18: Figure representing the BEM simulation for the NACA-6409 airfoil

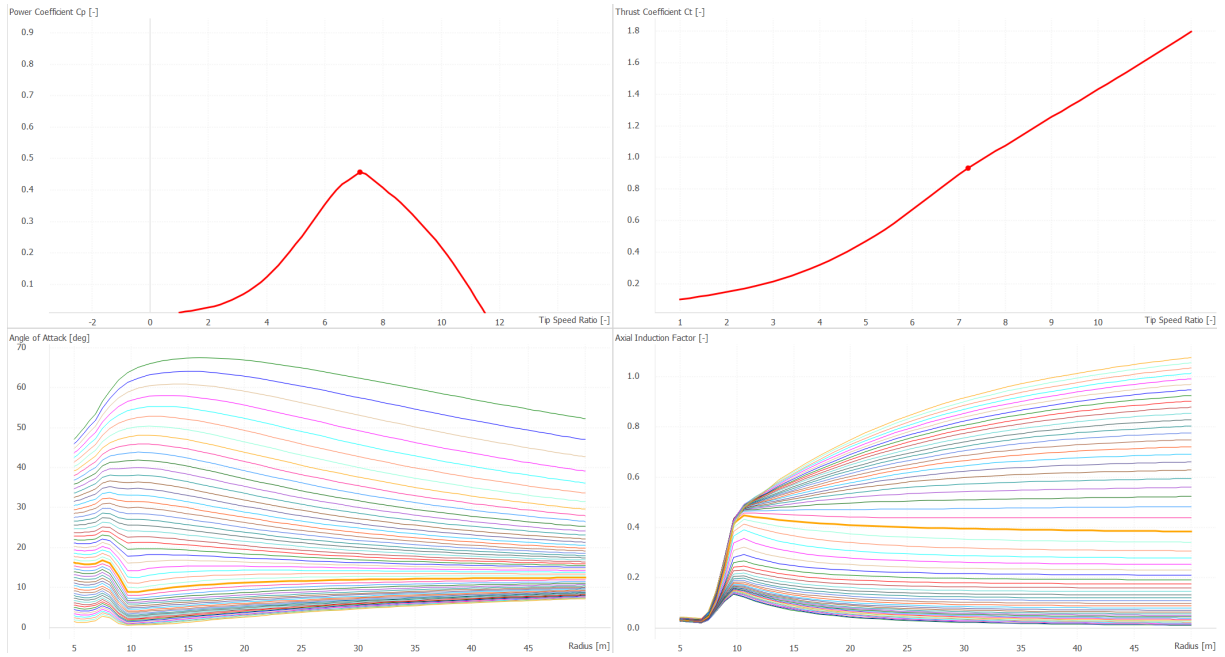


Figure 19: Figure representing the BEM simulation for the NACA-4412 airfoil

Power generated by each airfoil-based turbine is computed using:

$$P = \frac{1}{2}\rho AV^3 C_p$$

Table 9: Power output comparison.

Airfoil	Optimal AoA	C_p
NACA-6409	6.5	0.445
NACA-4412	6.5	0.459

- $V=7.52$ m/s
- $A=7854m^2$
- $\rho = 1.2$ kg/m³

$$P(6409) = 891.775Kw$$

$$P(4412) = 919.831Kw$$

$$PBetz = 1188.366Kw$$

Now computing the efficiency of each airfoil compared to Betz we find the following:

$$\eta_{\text{NACA-6409}} = 0.75042$$

$$\eta_{\text{NACA-4412}} = 0.77403$$

5 Optimization

By inspection the point having the lowest drag coefficient C_d was the angle of attack= 1 for the NACA-6409 airfoil and angle of attack= 0.5 for the NACA-4412 airfoil, this value was chosen by taking several factors into consideration such as the glide ratio and the pressure plot.

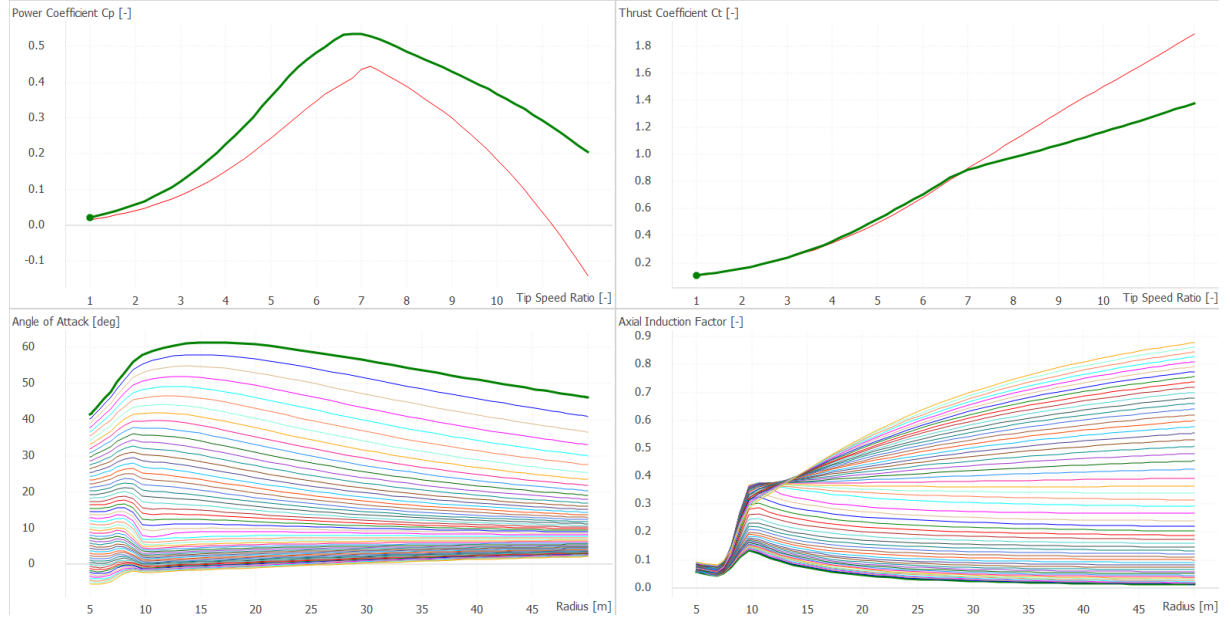


Figure 20: Figure showing the optimized cp and the old cp for the NACA-6409 airfoil

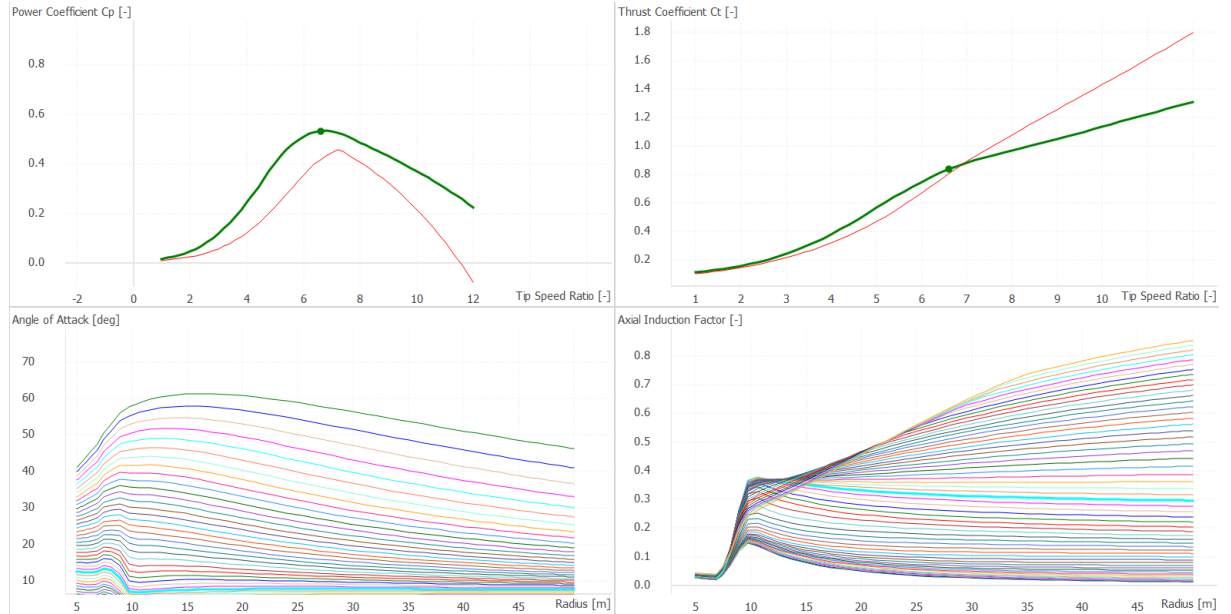


Figure 21: Figure showing the optimized cp and the old cp for the NACA-4412 airfoil

The new C_p 's are respectively:

- 0.536 for the NACA-6409 airfoil
- 0.533 for the NACA-4412 airfoil

Calculating the new power respectively:

$$P(6409) = 1074.183 Kw$$

$$P(4412) = 1068.126 Kw$$

Calculating the new efficiencies compared to Betz.

$$\eta_{\text{NACA-6409}} = 0.9045$$

$$\eta_{\text{NACA-4412}} = 0.8994$$

Feasibility and choice

From these results we consider the optimized design of the NACA-6409 to be more advantageous than the NACA-4412 that is because it had a better performance in terms of structure analysis and aerodynamic efficiency (optimized design), however when purchasing a wind turbine, it is important to consider both the cost and the quality of manufacturing. The total cost includes not only the turbine itself, but also the tower, foundation, electrical components, transportation, and installation. High-quality manufacturing ensures that the blades, hub, and structural components are produced with durable materials and precise engineering, which reduces maintenance needs and increases the turbine's lifespan. The reliability of the drivetrain, bearings, and control systems also depends heavily on manufacturing standards. Choosing a well-built turbine may have a higher upfront cost, but it typically provides better long-term performance, lower failure rates, and reduced operating expenses compared to cheaper, poorly manufactured alternatives.

6 Conclusion

6.1 Project Summary

This project's objective was to evaluate and compare the performance of two horizontal-axis wind turbines, set at Dahr El Baydar with a mean wind speed of 7.52 m/s, each with its own airfoil that were selected according to criteria coming from Lebanon's atmosphere. The methodology was based on using the Qblade software to analyze the glide ratio's (Cl/Cd) performance, the pressure distribution across the airfoils at different angles of attack, the turbine's power output, structural loading and coefficient of performance. Optimization on the coefficient of performance based on lowest drag for the highest power generated under real conditions. The findings collected let us reach the aerodynamic efficiency, mechanical stresses and the range of feasibility for an onshore wind turbine.

6.2 Airfoil Comparison

In terms of Aerodynamic performance NACA-6409 achieved remarkable performances, it had a higher cl/cd ratio and a slightly higher power output 1074 kw than the NACA-4412 airfoil 1068 kw. In terms of Robustness the NACA-6409 surprisingly endured a lower stress 58.87 MPA than NACA-4412 63.1 MPA exhibiting stronger suction and smoother flow distributions.

6.3 Recommendation

Regarding Lebanon's setting in terms of wind quality and unique terrain we would like to recommend a wind Turbine having the following properties:

- **hub-height:** 80m
- **blade length:** 45m
- **hub radius:** 5m
- **airfoil:** NACA-6409
- **Location:** Dahr el-Baidar

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